Soft calibration

Soft calibration nudges model outputs toward regional targets expressed as fractions of precipitation (water balance), per-stream-order channel sediment budgets, and per-region plant yields. The targets are user-supplied and the model adjusts a small fixed set of parameters internally until each component falls inside its target window. Soft calibration is run separately from a normal simulation; it loops over the relevant land-use regions and process components, updates parameters, and re-runs the affected balance routine.

Components

The soft codes file (`codes.sft`) is read by `calsoft_read_codes.f90` into `cal_codes` (`type soft_calibration_codes` in `calibration_data_module.f90`). Each switch enables one component of the soft calibration:

Code	Switch	Component	Driver routine
`hyd_hru`	`n`, `a`, `b`, `y`	HRU water balance by land use in each region. `a` calibrates all components (ET, surface runoff, lateral flow, percolation). `b` calibrates baseflow and total runoff only.	`calsoft_hyd.f90`, `calsoft_hyd_bfr.f90`
`hyd_hrul`	`y/n`	`hru_lte` water balance by land use in each region	`caltsoft_hyd.f90`
`plt`	`y/n`	Plant growth (LAI, harvest index) by region	`calsoft_plant.f90`
`sed`	`y/n`	Upland sediment yield by land use in each region	`calsoft_sed.f90`
`nut`	`y/n`	HRU nutrient balance by land use in each region	(gated in `calsoft_control.f90`)
`chsed`	`y/n`	Channel widening and bank accretion by stream order	(gated in `calsoft_control.f90`)
`chnut`	`y/n`	Channel nutrient balance by stream order	(gated in `calsoft_control.f90`)
`res`	`y/n`	Reservoir budgets by reservoir	(gated in `calsoft_control.f90`)

If any switch is non-default the master flag `cal_soft` is set to `"y"` and `calsoft_control` is called instead of (or alongside) the normal simulation.

HRU water balance

`calsoft_hyd.f90` is the full water-balance loop. It iterates over land-use regions, reads the regional water-balance targets (`water_balance.sft`) and the per-parameter perturbation envelope (`wb_parms.sft`), and adjusts the active parameter set until each component is within tolerance. The components and their helper routines are:

- PET, `calsoft_hyd_bfr_pet.f90`
- ET, `calsoft_hyd_bfr_et.f90`
- Surface runoff, `calsoft_hyd_bfr_surq.f90`
- Lateral flow, `calsoft_hyd_bfr_latq.f90`
- Percolation, `calsoft_hyd_bfr_perc.f90`

The shared baseflow ratio loop is `calsoft_hyd_bfr.f90`.

Adjustable parameters for the upland water balance are read into `ls_prms` (`type soft_calib_parms`). Each entry carries a parameter name, the change type (`absval`, `abschg`, or `pctchg`), and `neg`/`pos` perturbation limits with absolute floor/ceiling values.

Channel sediment

`ch_sed_budget.sft` carries the per-stream-order sediment budget targets. `ch_sed_parms.sft` carries the perturbation envelope for channel sediment parameters, read into `ch_prms`. The channel sediment block is gated in `calsoft_control.f90` but the call to a `calsoft_chsed` driver is commented out in the current source; only the per-order accounting loop runs.

Plant growth

`plant_parms.sft` defines the plant parameter envelope (LAI potential, harvest index) and `plant_gro.sft` carries the per-region plant target table. The driver is `calsoft_plant.f90`; `calsoft_plant_zero.f90` zeros the running statistics between iterations. After soft calibration writes the new plant parameter values, `pl_write_parms_cal` dumps a fresh `plants.plt` and a `hydrology-cal.hyd` snapshot.

Outputs

`calsoft_control.f90` writes:

- `hydrology-cal.hyd` (unit 5001) with per-HRU hydrology parameters after the run.
- A channel sediment summary (unit 5000) when `chsed = y`.

These outputs are intended to be copied back into the model inputs to lock in the soft-calibrated values.

Input files

File	Read by	Purpose
`codes.sft`	`calsoft_read_codes.f90`	Component switches.
`wb_parms.sft`	(read in `calsoft_init`)	Upland water-balance parameter envelope.
`water_balance.sft`	(read in `calsoft_init`)	Regional water-balance targets (SR, LF, PC, ET, TF as fractions of precipitation).
`ch_sed_budget.sft`	(read in `calsoft_init`)	Channel sediment budget targets by stream order.
`ch_sed_parms.sft`	(read in `calsoft_init`)	Channel sediment parameter envelope.
`plant_parms.sft`	(read in `calsoft_init`)	Plant parameter envelope.
`plant_gro.sft`	(read in `calsoft_init`)	Per-region plant yield targets.

All `*.sft` slots are declared in `input_file_module.f90`, `type input_chg`. Set the slot to `null` (or remove the file) to skip a component.

Related pages

- Hard calibration. Use after soft calibration to fit remaining parameters to observed time series.
- Parameter change files. `cal_parms.cal` defines the parameter universe and bounds shared with hard calibration.